



JSS MAHAVIDYAPEETHA
JSS University, Noida
CIE-1 (AY 2026-27: Odd Semester)



Semester: III
Subject Name: Machine Learning Techniques
Date: 16/09/2026 | Time: 1 Hour

Course: MCA
Subject Code: 25MCA310
Max. Marks: 20

COURSE OUTCOMES:

- CO1: Understand the concepts of Artificial Intelligence and Machine Learning, including types of learning, data pre processing, and the data processing cycle.
CO2: Apply supervised machine learning algorithms such as regression, classification, and ensemble methods to solve real-world problems.
CO3: Analyze and implement unsupervised learning techniques including clustering, association rule mining, and probabilistic models.
CO4: Analyze reinforcement learning frameworks including Markov Decision Processes, Q-learning, and temporal difference learning for decision-making problems.
CO5: Design and apply neural network and deep learning architectures such as CNNs and RNNs for real-world applications.

PART A: ANSWER ALL THE QUESTIONS

Each question carries 1 mark

4Qx1=04

Q	QUESTIONS	BL	CO
1	A Define Artificial Intelligence (AI).	1	CO1
	B List any two types of Machine learning.	1	CO1
	C State the purpose of Data Preprocessing in Machine Learning.	2	CO2
	D Give one real life example of Regression and Classification respectively.	2	CO2

PART B: ANSWER ALL THE QUESTIONS

Each question carries 4 marks

3QX4=12

Q	QUESTIONS	BL	CO
2	a Explain the different types of Machine learning (Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning) with one suitable example for each.	2	CO1
	OR		
3	b Illustrate the Data Processing Cycle with a neat diagram and briefly explain each stage involved.	3	CO1
	OR		
3	a Analyze the key differences between Artificial Intelligence and Data Science, with suitable examples.	4	CO4
	OR		
4	b Explain Linear Regression and Multiple Regression. Mention one difference between them.	2	CO1
	OR		
4	a Explain the working of K-Nearest Neighbour (KNN) classifier with a suitable example and list a few limitations of KNN.	2	CO2
	OR		
4	b A KNN classifier uses $k = 3$. For a new point $P = (3, 3)$, the three nearest training points are $(2, 2) \rightarrow$ Class A, $(3, 2) \rightarrow$ Class A, and $(5, 4) \rightarrow$ Class B. Apply KNN and determine the class of P. Give the reason for your answer.	3	CO2

PART C: ANSWER ANY 1 QUESTION

Each question carries 4 marks

1Qx4=4

Q	QUESTIONS	BL	CO
5	Write short notes on the following (2 x 2): a) SVM b) Decision Tree	2	CO2
	OR		

6	Toss	Venue	Outlook	Result	3	CO2
	Won	Mumbai	Overcast	Won		
	Lost	Chennai	Sunny	Won		
	Won	Kolkata	Sunny	Won		
	Won	Chennai	Sunny	Won		
	Lost	Mumbai	Sunny	Lost		
	Won	Chennai	Overcast	Lost		
	Won	Kolkata	Overcast	Lost		
	Won	Mumbai	Sunny	Won		

Apply Naïve Bayes Classifier to predict whether the person will Win or Lose, on given condition: {Toss = Lost, Venue = Mumbai, Outlook = Sunny}